PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Local anesthetics combined with vasoconstrictors in patients with cardiovascular disease undergoing dental procedures: Systematic review and meta-analysis
AUTHORS	Guimaraes, Caio Lopes, Luciane; Bergamaschi, Cristiane; Ramacciato, Juliana; Silva, Marcus; Araújo, Jimmy; de Andrade, Natalia Karol; Motta, Rogério

VERSION 1 – REVIEW

REVIEWER	Brown, RS Howard University , Comprehensive Dentistry
REVIEW RETURNED	04-Dec-2020

GENERAL COMMENTS	Intro: Issue taken with the following: The maximum recommended dose for epinephrine is 40 µg per procedure (2, 6, 9). There are other valid guidelines which recommend at least 60 micrograms of epinephrine per procedure! Furthermore, it should be noted that as epinephrine is a vasoconstrictive agent and injections are given into peripheral tissues, that epinephrine is released gradually into the circulation and furthermore the half life of epinephrine (once it is within the circulation) is less than two minutes! Therefore, safe dosage is probably well beyond 60 mcg of epinephrine per procedure. It is imperative that both the ying and the yang of recommended dosage be provided both the lesser and the greater recommendations absolutely need to be provided. There are both textbook and publication references regarding such available.

REVIEWER	Skaar, Daniel D. Univ Minnesota
REVIEW RETURNED	15-Jan-2021

GENERAL COMMENTS	Although I thank the authors for their efforts, I simply do not believe the manuscript merits publication. Despite the attempted application of meta analysis, the results and conclusions do not advance the literature on the subject. In the end, the manuscript mainly summarizes the results of several past articles and admits the low quality of evidence prevents meaningful conclusions. The conclusion that LA with vasoconstrictor reduces bleeding and and pain is not novel. Discussion of CVD effects is limited by the small number of studies and their variability in defining and reporting CVD. Safety, as defined in the abstract, seems to change during the manuscript. I am not sure pain and bleeding, in this context, would be universally considered safety issues unless patient risk and the extent was defined. As the authors found and point out the type and concentrations of LA and vasoconstrictor vary across

	studies, definitions and measuring of CVD varies and is incomplete, and the number of studies and patients ultimately left in the meta analysis are too small to provide meaningful results. I do agree the low quality of results indicates further study needs.

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1
Dr. RS Brown, Howard University
1) Issue taken with the following: The maximum recommended dose for epinephrine is 40 µg per procedure (2, 6, 9). There are other valid guidelines which recommend at least 60 micrograms of epinephrine per procedure! Furthermore, it should be noted that as epinephrine is a vasoconstrictive agent and injections are given into peripheral tissues, that epinephrine is released gradually into the circulation and furthermore the half life of epinephrine (once it is within the circulation) is less than two minutes! Therefore, safe dosage is probably well beyond 60 mcg of epinephrine per procedure. It is imperative that both the ying and the yang of recommended dosage be provided both the lesser and the greater recommendations absolutely need to be provided. There are both textbook and publication references regarding such available.
We have edited the text as requested. This statement was included in the Introduction section of the manuscript:
"A few decades ago, the maximum dose of epinephrine in patients with cardiovascular disease was set at up to 200 µg (16). However, over the years several studies and guidelines on this subject have been published and suggested that 36 to 54 µg of epinephrine appear to be tolerated in most patients with hypertension or other cardiovascular diseases (2,17)."

Dear Reviewer,
We would like to thank you very much for your interesting comments. We have edited and corrected the paper according to your suggestions. We also highlighted the changes within our manuscript using colored text.

Reviewer: 2
Dr. Daniel D. Skaar, Univ. Minnesota

Comments to the Author:
Although I thank the authors for their efforts, I simply do not believe the manuscript merits publication. Despite the attempted application of meta-analysis, the results and conclusions do not advance the literature on the subject. In the end, the manuscript mainly summarizes the results of several past articles and admits the low quality of evidence prevents meaningful conclusions. The conclusion that LA with vasoconstrictor reduces bleeding and pain is not novel. Discussion of CVD effects is limited by the small number of studies and their variability in defining and reporting CVD.
Thank you very much for your comments. There is a gap in the literature regarding systematic reviews considering the randomized clinical studies about safety of the use of local anesthetics with vasoconstrictors (especially in patients with cardiovascular diseases). As mentioned in the introduction of the manuscript, the existing systematic reviews have some limitations that can interfere with the analysis of the level of evidence of the findings.
In this context, this study was carried out adopting strict criteria in relation to the study designs and inclusion criteria in order to contribute with information on this topic, as well as to point out the possible need for further studies. There are several types of cardiovascular diseases that can impact the patient with different comorbidities. This can make it difficult to conduct clinical studies about the safety of LA in these systemic patient profiles. Thus, this may explain the limited number of CVD found in the studies included in the present study. Considering this scenario, it is also the role of the

present systematic review to point out the need for further studies in this regard. Regarding the conclusion, we agreed with the comment and corrected it in the manuscript.

Safety, as defined in the abstract, seems to change during the manuscript.
Sorry for this inconvenience. We have edited the text as requested in the Abstract and in the Methods section of the manuscript.

I am not sure pain and bleeding, in this context, would be universally considered safety issues unless patient risk and the extent was defined.
Thank you for your recommendation. We consider bleeding as one of the outcomes due to the comparison of local anesthetics with and without vasoconstrictor. In addition, in the planning of the systematic review this parameter was also considered due to the possibility of inclusion of patients using platelet aggregation inhibitors (such as aspirin) or anticoagulants in the studies found. Pain was considered as primary outcome due to the fact that some clinical studies considered it as one of the main assessed outcomes. One possibility for this occurrence is the systemic release of endogenous catecholamines in pain and stressful conditions, which can lead to significant changes in heart rate and blood pressure of patients with CVD.

As the authors found and point out the type and concentrations of LA and vasoconstrictor vary across studies, definitions and measuring of CVD varies and is incomplete, and the number of studies and patients ultimately left in the meta-analysis are too small to provide meaningful results. I do agree the low quality of results indicates further study needs.
We agree with your recommendation. We have inserted this information in sections “Strengths and limitations of this study.” and “Implications for clinical practice and research”.

Dear Reviewer,
We would like to thank you very much for your interesting comments. We have edited and corrected the paper according to your comments. We also highlighted the changes within our manuscript using colored text.

Editors' Comments to Author:
You looked at the safety of LA with vasoconstrictor and the risk of cardiovascular events in patients with CVD – can you better define what you mean by safety. The only primary outcomes looked at were pain, bleeding and arrhythmias. The Conclusion in the Abstract and main body of the paper need to be toned down, and please explain the low quality of the evidence more clearly.
Thank you very much for your comments. We have edited the text about safety and primary outcomes in the Abstract and in the Methods section of the manuscript. In general, the quality of the evidence was considered low for the outcomes evaluated mainly due to the risk of bias, such as lack of blinding, concealment of the allocation not clearly described and problems due to the high heterogeneity between clinical trials and the inaccuracy of the findings. This statement was included in the manuscript.

There was no statistical difference between the use of LA with vasoconstrictor and without vasoconstrictor in patients with CVDs for the assessed outcomes.
This was one of the observations found in the study's meta-analysis. In this context, the use of the vasoconstrictor could be an interesting alternative for some clinical procedures to provide longer anesthesia.

6 of the included studies were published in Brazil. What are the implications for bias if any?
As described in the study methodology, the search strategy was carried out in different international databases. We have included this context in the manuscript.

The dates of the search are conflicting (Dec 2019 in the Abstract – but Jan 2020 in the paper) - please correct.
Sorry for this inconvenience. We have edited the text as requested.

Please revise the term ‘anaesthetic salt’ as this isn’t widely used. Better to say ‘local anaesthetic’
Sorry for this inconvenience. We have edited the text as requested.